Lecture 28

Integrals in geometry and physics

Uses.

Summary 2: Every nonempty real set bounded above has a supremum.
Summary 8: Elementary differentiation and the chain rule hold.
Summary 9: The scalar mean value theorem applies to a function continuous on a closed interval and differentiable inside it.
Summary 12: The Euclidean norm, its triangle inequality and its reverse triangle inequality hold in $\mathbb{R}^n$.
Summary 13: A constant force does work equal to its dot product with displacement.
Summary 14: Polar coordinates satisfy $x = r\cos\varphi$, $y = r\sin\varphi$. The ellipse $r = p/(1+\varepsilon\cos\varphi)$ has centre $h = -\varepsilon p/(1-\varepsilon^2)$ and semiaxes $a = p/(1-\varepsilon^2)$, $b = p/\sqrt{1-\varepsilon^2}$.
Summary 25: For continuous functions on a closed finite interval, Darboux gaps and arbitrary-tag Riemann-sum errors tend to zero with the mesh. Uniform continuity gives one bound over the whole interval.
Summary 26: Continuous functions are integrable; primitives evaluate their integrals. Definite substitution transforms endpoints and derivative factors.

Objectives. *A partition divides a geometric or physical quantity into finite pieces. Their sums identify the integrand before any antiderivative is chosen. We apply this method to area, volume and length, then to work and to the mass and moments of one nonuniform rod.*

By the end of this lecture you will be able to:

- Set up Cartesian and polar area integrals with their geometric bounds.
- Derive volume formulas for solids of revolution by finite-slice bounds.
- Prove the length formula for continuously differentiable parametrised curves.
- Compute signed work, rod mass, centre of mass and moment of inertia from their densities.

28.1 From finite pieces to an integral

For a continuous function $w : [a, b] \to \mathbb{R}$, Lecture 25 shows that

$$\sum_i w(\xi_i)\Delta x_i \longrightarrow \int_a^b w(x)\,\mathrm{d}x \qquad \text{as the partition mesh tends to zero,}$$

for arbitrary tags $\xi_i \in [x_{i-1}, x_i]$. A model must explain why its pieces have this limiting sum. The notation $\mathrm{d}A = w(x)\,\mathrm{d}x$ records the resulting integral density. It does not assert that a finite strip has exactly the area $w(x)\Delta x$ for every tag. We use elementary geometric area

and volume for rectangles, circular sectors and cylinders, together with finite additivity and monotonicity under inclusion.

28.2 Area between graphs and in polar coordinates

Let $f, g : [a, b] \to \mathbb{R}$ be continuous with $g \le f$. The region between their graphs consists of points satisfying $a \le x \le b$ and $g(x) \le y \le f(x)$. On each partition interval, let m_f, M_f and m_g, M_g be the respective minimum and maximum values. An outer rectangle has height $M_f - m_g$. An inner rectangle, when its height is positive, has height $m_f - M_g$. Use zero inner height otherwise. The difference between these two heights is at most $(M_f - m_f) + (M_g - m_g)$. After multiplying by widths and summing, the total area gap is at most the sum of the two Darboux gaps, which tends to zero. These inner and outer rectangles therefore define one common area. The tagged heights $f(\xi_i) - g(\xi_i)$ lie between the same bounds, so

$$A = \int_a^b (f(x) - g(x))\,\mathrm{d}x. \tag{28.1}$$

If two graphs cross finitely many times, split at those intersections. For graphs that may cross, the region consists of the points with $a \le x \le b$ and y between $\min(f(x), g(x))$ and $\max(f(x), g(x))$. Apply the area formula to these lower and upper graphs. Both are continuous because

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \qquad \min(f, g) = \frac{f + g - |f - g|}{2}.$$

Their difference is $|f - g|$, so the area is

$$A = \int_a^b |f(x) - g(x)|\,\mathrm{d}x,$$

including when there are infinitely many crossings. The signed integral of $f - g$ alone can cancel areas from different pieces.

Figure 28.1 contrasts vertical strips with angular sectors. The height and radius must come from the region being measured. In the angular panel, $r(\varphi)$ is the distance from the origin to the outer boundary along the ray of angle φ.

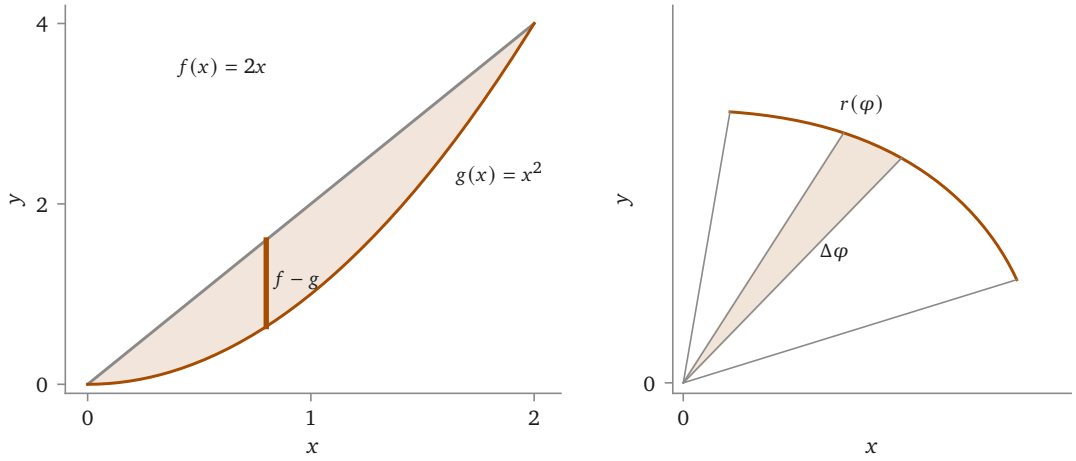


Figure 28.1: A vertical piece uses the difference of graph heights. The angular panel shows a radial piece between two rays, with its variable outer radius.

Example 28.1 (Area requires the upper graph). Find the area between $y = x^2$ and $y = 2x$. Their intersections solve $x^2 = 2x$, hence $x = 0$ or $x = 2$. On $[0, 2]$, the difference $2x - x^2 = x(2 - x)$ is nonnegative. Therefore

$$A = \int_0^2 (2x - x^2)\,\mathrm{d}x = \left[x^2 - \frac{x^3}{3}\right]_0^2 = \frac{4}{3}.$$

For a polar region, suppose a continuous radius $r : [\alpha, \beta] \to [0, \infty)$ bounds every ray once, with $0 < \beta - \alpha \le 2\pi$. The region has $0 \le \rho \le r(\varphi)$ and $\alpha \le \varphi \le \beta$ in the polar coordinates of Lecture 14. For a constant radius R, a sector of angle $\Delta\varphi$ has area $R^2\Delta\varphi/2$, because radians measure that fraction of a full circle. On each angular subinterval, sectors with the minimum and maximum radii bound the region. Their areas are lower and upper sums for $r^2/2$. Continuity makes their gap tend to zero, proving

$$A = \frac{1}{2}\int_\alpha^\beta r(\varphi)^2\,\mathrm{d}\varphi. \tag{28.2}$$

The angular restriction prevents counting a region repeatedly. The common boundary ray of a full turn has zero area and causes no duplication of area.

A parametric substitution also computes Cartesian area. Suppose the full vertical height is $H(x)$ and $x = X(t)$ maps $[u, v]$ increasingly onto $[a, b]$. If H is continuous on a neighbourhood of $[a, b]$ and X is continuously differentiable on a neighbourhood of $[u, v]$, Lecture 26 gives $A = \int_u^v H(X(t))X'(t)\,\mathrm{d}t$. If H is only relatively continuous, extend it constantly beyond its two endpoints first; the extension is continuous there as in Sheet 26. A decreasing parametrisation requires reversed endpoints to keep the area positive.

Example 28.2 (The running ellipse). Compute the area of Lecture 14's ellipse with centre $(h, 0)$ and semiaxes $a \ge b > 0$. Its upper and lower heights are $\pm b\sqrt{1 - (x - h)^2/a^2}$ on $[h - a, h + a]$. Set $x = h - a\cos t$, with $0 \le t \le \pi$. The substitution is increasing, and the full

height becomes $2b\sin t$. Since $\mathrm{d}x = a\sin t\,\mathrm{d}t$, the finite substitution theorem gives

$$A = 2ab\int_0^{\pi}\sin^2 t\,\mathrm{d}t = ab\int_0^{\pi}(1-\cos 2t)\,\mathrm{d}t = \pi ab.$$

For the polar data p and ε from Lecture 14, this is $\pi p^2/(1-\varepsilon^2)^{3/2}$.

28.3 Volume by cross-sections

A *cross-section* at x is the intersection of a solid with the plane perpendicular to the x axis there. If its area is $A(x)$, the slicing method forms sums $\sum_i A(\xi_i)\Delta x_i$. When A is continuous, these converge to $\int_a^b A(x)\,\mathrm{d}x$. To identify this limit with geometric volume, the slices must bound the solid by finite inner and outer pieces. We now supply that justification for the solids of revolution used here.

Let $f:[a,b]\to[0,\infty)$ be continuous. Rotating the region under f about the x axis gives discs with radii $f(x)$. On a partition interval, let m_i and M_i be the minimum and maximum radii. Cylinders of those radii bound the corresponding solid piece, and their volumes are $\pi m_i^2\Delta x_i$ and $\pi M_i^2\Delta x_i$. These are lower and upper sums for the continuous function πf^2. Their gap tends to zero, so the common volume is

$$V = \pi\int_a^b f(x)^2\,\mathrm{d}x. \tag{28.3}$$

For continuous inner and outer radii $0\le r(x)\le R(x)$, annular cylinders give a direct bound. On each partition interval, denote the minimum and maximum of each radius by $r_{\min}, r_{\max}$ and $R_{\min}, R_{\max}$. The outer annular cylinder has squared-radius difference $R_{\max}^2 - r_{\min}^2$. The inner one has difference $\max(R_{\min}^2 - r_{\max}^2, 0)$; zero means an empty inner piece. The gap between these differences is at most

$$(R_{\max}^2 - R_{\min}^2) + (r_{\max}^2 - r_{\min}^2).$$

Multiplying by $\pi\Delta x_i$ and summing gives a gap tending to zero, since both squared radii are continuous. The tagged annular areas lie between the same bounds. This proves the *washer* formula $V = \pi\int_a^b (R^2 - r^2)\,\mathrm{d}x$, keeping both squared radii.

For a ball of radius $R>0$, the section at $x\in[-R,R]$ is a disc with squared radius R^2-x^2. The formula gives

$$V = \pi\int_{-R}^{R}(R^2-x^2)\,\mathrm{d}x = \pi\left[R^2x - \frac{x^3}{3}\right]_{-R}^{R} = \frac{4\pi R^3}{3}.$$

For a different example, washers with outer radius $2-x$ and inner radius x on $[0,1]$ have volume $\pi\int_0^1((2-x)^2 - x^2)\,\mathrm{d}x = 2\pi$. The inequalities between the radii hold throughout that interval.

28.4 Arc length

A *parametrised curve* is a continuous map $\boldsymbol{r}:[a,b]\to\mathbb{R}^d$, with $d=2$ or $d=3$ here. A partition gives its *polygonal length* $L(P) = \sum_i \|\boldsymbol{r}(t_i) - \boldsymbol{r}(t_{i-1})\|$. The *arc length* is the supremum of these lengths over all partitions, provided that supremum is finite. It counts repeated traversals of a path. We call a parametrisation *continuously differentiable*, or C^1, when its component functions have continuous derivatives on a neighbourhood of $[a,b]$. Its derivative $\boldsymbol{r}'$ is the vector of those derivatives.

Theorem 28.3 (Length of a C^1 parametrised curve)**.** *For such a parametrisation, the arc length exists and equals*

$$L = \int_a^b \|\boldsymbol{r}'(t)\| \, \mathrm{d}t.$$

The derivative is allowed to vanish.

Proof. Fix $\eta > 0$. Uniform continuity of the finitely many component derivatives supplies one mesh size for which each derivative changes by less than η within one partition interval. For every component j, the scalar mean value theorem gives a point $c_{ij} \in (t_{i-1}, t_i)$ with

$$r_j(t_i) - r_j(t_{i-1}) = r_j'(c_{ij})\Delta t_i.$$

Choose any common tag τ_i in that interval. The points c_{ij} may differ between components; uniform continuity still gives

$$\|\boldsymbol{r}(t_i) - \boldsymbol{r}(t_{i-1}) - \boldsymbol{r}'(\tau_i)\Delta t_i\| \le \sqrt{d}\,\eta\Delta t_i.$$

The reverse triangle inequality, summed over all subintervals, gives

$$\left| L(P) - \sum_i \|\boldsymbol{r}'(\tau_i)\| \, \Delta t_i \right| \le \sqrt{d}\,\eta(b-a).$$

The sum inside the absolute value tends to the integral of the continuous function $\|\boldsymbol{r}'\|$. Thus polygonal lengths tend to that integral as the mesh tends to zero.

Refinement cannot decrease polygonal length, by the triangle inequality. Given any fixed partition P, refine it with partitions of arbitrarily small mesh. Their lengths are at least $L(P)$ and tend to the integral, so $L(P)$ is at most that integral. Conversely, arbitrarily fine polygonal lengths approach it, so their supremum is at least the integral. Both inequalities identify the supremum and prove finiteness. ∎

For a graph, use $\boldsymbol{r}(x) = (x, f(x))$ to obtain $L = \int_a^b \sqrt{1 + f'(x)^2}\,\mathrm{d}x$. The notation $\mathrm{d}s = \|\boldsymbol{r}'(t)\| \, \mathrm{d}t$ is the length density. Sheet 28 derives its polar form, keeping both radial and angular changes.

Figure 28.2 shows a graph and an inscribed polygon.

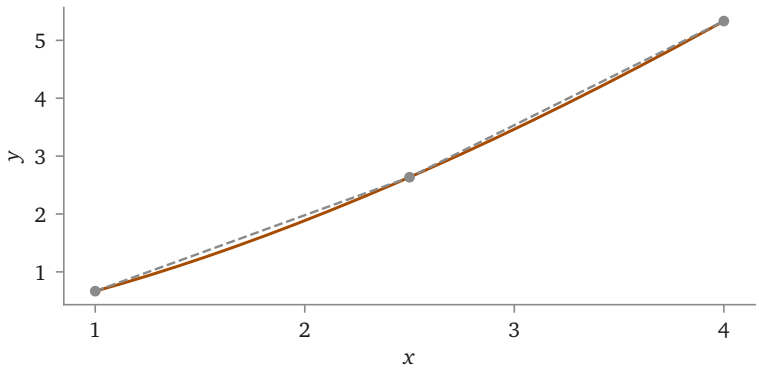


Figure 28.2: Polygonal approximation to $y = (2/3)x^{3/2}$ for $1 \le x \le 4$. The length theorem concerns all sufficiently fine partitions, not only equal steps.

Example 28.4 (Graph length and ellipse perimeter). For $f(x) = (2/3)x^{3/2}$ on $[1, 4]$, differentiation gives $f'(x) = \sqrt{x}$. Thus its graph length is

$$L = \int_1^4 \sqrt{1 + x}\,\mathrm{d}x = \left[\frac{2}{3}(1+x)^{3/2}\right]_1^4 = \frac{2}{3}(5\sqrt{5} - 2\sqrt{2}).$$

For the ellipse, use the parametrisation and derivative

$$\boldsymbol{r}(t) = (h + a\cos t, b\sin t), \qquad \boldsymbol{r}'(t) = (-a\sin t, b\cos t), \quad 0 \le t \le 2\pi.$$

Symmetry of the squared sine and cosine, followed by $u = \pi/2 - t$ in a quarter, gives its perimeter

$$P = 4\int_0^{\pi/2} \sqrt{a^2\sin^2 t + b^2\cos^2 t}\,\mathrm{d}t = 4a\int_0^{\pi/2} \sqrt{1 - \varepsilon^2\sin^2 u}\,\mathrm{d}u, \qquad \varepsilon^2 = 1 - b^2/a^2.$$

The nonnegative argument $\varepsilon = \sqrt{1 - b^2/a^2}$ equals the parameter of the running polar ellipse in Lecture 14, since its semiaxes satisfy $b^2/a^2 = 1 - \varepsilon^2$. The last integral defines the *complete elliptic integral of the second kind* $E(\varepsilon)$. Its argument is called the *modulus*. No elementary evaluation is asserted. For $a = b$, its integrand is 1 and the perimeter is $2\pi a$.

28.5 Variable-force work

Consider motion along a straight coordinate axis. Let $F(x)$ be the signed component of force along increasing x, continuous between the endpoints. Over a short displacement, replacing the force by $F(\xi_i)$ gives the constant-force work $F(\xi_i)\Delta x_i$ from Lecture 13. The *work* of the variable force is the limit of these sums:

$$W_{a\to b} = \int_a^b F(x)\,\mathrm{d}x.$$

The reversed-endpoint convention handles motion towards smaller x. Work is signed: a force opposed to displacement contributes negatively. If force is measured in newtons and displacement in metres, work is in joules, where one joule is one newton metre.

A Hooke spring has stiffness $k > 0$ and restoring force $-kx$ at extension $x \geq 0$ from its natural length. In a slow stretch, neglect changes of kinetic energy and balance that force by an external force kx. The external work from zero extension to $X \geq 0$ is

$$W_{\text{ext}} = \int_0^X kx \, \mathrm{d}x = \frac{kX^2}{2}.$$

The spring's restoring force does the negative work $-kX^2/2$ on the same stretch. Returning from X to zero reverses each signed work.

28.6 Mass and moments of a rod

Let a straight rod occupy $[0, L]$, with $L > 0$ and continuous linear mass density $\lambda : [0, L] \to [0, \infty)$ that is not identically zero. The density is mass per unit length. The mass is the limit of piece masses $\lambda(\xi_i)\Delta x_i$, so $m = \int_0^L \lambda(x) \, \mathrm{d}x$. This is positive: continuity at a point of positive density gives a positive lower bound on some interval of positive length. The *first moment about zero* weights each mass by its coordinate. The *centre of mass coordinate* is that first moment divided by total mass:

$$Q_0 = \int_0^L x\lambda(x) \, \mathrm{d}x, \qquad \bar{x} = \frac{Q_0}{m}. \tag{28.4}$$

These formulas are limits of the finite point-mass formulas on Sheet 12: the numerator and denominator Riemann sums converge, and $m > 0$ permits division. Order gives $0 \leq Q_0 \leq Lm$, so the centre lies in the rod. In fact, $\int_0^L (x - \bar{x})\lambda(x) \, \mathrm{d}x = Q_0 - \bar{x}m = 0$.

The *moment of inertia* about a line perpendicular to the rod at coordinate $c \in \mathbb{R}$ weights each mass by its squared distance from that line:

$$I_c = \int_0^L (x - c)^2 \lambda(x) \, \mathrm{d}x.$$

It is nonnegative and has units of mass times length squared. Expand $x - c = (x - \bar{x}) + (\bar{x} - c)$ inside the square and use linearity. The cross term integrates to zero by the preceding first-moment identity. This proves the *parallel-axis formula*

$$I_c = I_{\bar{x}} + m(c - \bar{x})^2. \tag{28.5}$$

Among these parallel axes, the one through the centre of mass minimises inertia.

Fix $\lambda_0 > 0$. Figure 28.3 shows the rod density $\lambda_0(1+2x/L)$. Its centre of mass is displaced towards the denser end.

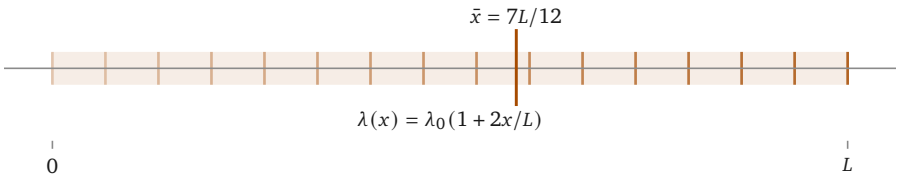


Figure 28.3: For density $\lambda_0(1+2x/L)$, darker marks indicate larger mass per unit length. The centre is at $7L/12$, to the right of the midpoint.

Example 28.5 (One rod, three weighted integrals). Let $\lambda(x) = \lambda_0(1+2x/L)$ for $0 \le x \le L$, with $\lambda_0 > 0$. Compute its mass, centre of mass, and inertia about the left endpoint and about its centre of mass. The defining integrals give

$$m = \lambda_0 \left[x + \frac{x^2}{L}\right]_0^L = 2\lambda_0 L,$$
$$Q_0 = \lambda_0 \left[\frac{x^2}{2} + \frac{2x^3}{3L}\right]_0^L = \frac{7}{6}\lambda_0 L^2,$$
$$I_0 = \lambda_0 \left[\frac{x^3}{3} + \frac{x^4}{2L}\right]_0^L = \frac{5}{6}\lambda_0 L^3.$$

Consequently $\bar{x} = Q_0/m = 7L/12$. The parallel-axis formula then gives

$$I_{\bar{x}} = I_0 - m\bar{x}^2 = \left(\frac{5}{6} - 2\frac{49}{144}\right)\lambda_0 L^3 = \frac{11}{72}\lambda_0 L^3.$$

Summary 28.

- Continuous graph heights give $A = \int (f-g)$ where $f \ge g$. A single polar sweep with continuous $r \ge 0$ gives $A = \frac{1}{2}\int r^2$. Both follow by squeezing finite geometric pieces.
- Discs of continuous radius give $V = \pi \int f^2$; continuous washers give $V = \pi \int (R^2 - r^2)$ with $0 \le r \le R$.
- Theorem 28.3: a C^1 parametrised curve has length $\int \|\boldsymbol{r}'\|$, even when its derivative vanishes. Repeated traversals are counted.
- Continuous force components give signed work $\int F\,\mathrm{d}x$. For a slow Hooke-spring stretch, external work is $kX^2/2$.
- A nonnegative continuous rod density with positive total mass gives $m = \int \lambda$, $\bar{x} = (\int x\lambda)/m$ and $I_c = \int (x-c)^2\lambda$. The parallel-axis formula is Eq. (28.5).

Sheet 28 develops polar length, disc inertia and further geometric and physical setups. Plane-region centres of mass return in Lecture 38. Lecture 29 asks when an infinite sum converges.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§6.1–6.2, 6.4, 8.1, 8.3 and 10.4.